



SOUR MILK STUDIOS · PLAYER AND CLASSROOM GUIDE

Sour Milk Logistics

Tutorial and Finance Study Guide

The full guided tutorial, the rules the game runs on, and the finance course material each part of the game puts into practice.

Part 1 · Play

Getting started, all 13 tutorial steps, and the objectives that follow.

Part 2 · Learn

Ten finance modules, each with the game rule, the theory, a formula and a worked example.

Part 3 · Compete

How to run an online team match as a class session.

Part 4 · Practise

A course map, twelve practice questions and the answer key.

Play free in a browser: logistics.sourmilkstudios.com

Based on the game as published on 5 October 2026. Contact: sourmilkstudios1@gmail.com · Instagram [@sourmilkstudios_official](https://www.instagram.com/sourmilkstudios_official)

How to use this guide

Sour Milk Logistics puts you in charge of a small shipping company inside a working world economy of ten countries and 79 companies. This guide walks you through the tutorial, then shows what each part of the game teaches a finance student.

For players

Read Part 1 next to the game. It follows the in-game tutorial step by step and adds the rules the guide character does not have time to explain.

For students

Play the tutorial first, then work through the modules in Part 2 in order. Each one ends with something to try in the game. Part 4 has practice questions with answers.

For lecturers and student organisations

Use the course map in Part 4 to pick the modules that match your syllabus. Part 3 explains how to host an online match so that teams compete in one shared economy.

About the numbers

Every rule, rate and price in this guide describes the game, not real markets. Ship and research prices are the starting values; the game's price level rises over time.

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1.1 Getting started

The game is free and runs in a web browser at logistics.sourmilkstudios.com. Nothing needs to be installed. Before a new game you make three choices on the title screen.



In the game. The title screen: Play online, starting capital, the tutorial switch and the two modes.

CHOICE	OPTIONS	WHAT TO PICK
Mode	Normal or Pro	Normal first; Pro for the full financial markets.
Starting capital	Founder \$12.48M · Investor \$1B · Tycoon \$60B	Founder is the intended game. Tycoon can buy control of any company on day one.
Guided tutorial	Play the tutorial or skip it	Play it. It takes about two minutes.

MODE	WHAT IT INCLUDES
Normal	10 countries · Trade and logistics · Warehouses and factories · Contracts · Simplified stock market (shares for dividends and gains)
Pro	79 public companies you can inspect · Detailed company management · Advanced stock and bond market · Economic news and global events · Central banks, interest rates, inflation · Full supply chain simulation

The screens

The left menu holds ten screens. The gear at the top right holds saving, the length of a day and the tutorial.

SCREEN	WHAT YOU DO THERE	SCREEN	WHAT YOU DO THERE
World	The live map, your objectives and anything that needs attention	Factories	Found your own producer: product, country, size
Companies	Open any company and look inside it	Fleet	Your ships, repairs and the shipyard
Market	Commodity prices and stocks; in Pro mode also bonds and central banks	Research	Six projects that improve the fleet, contracts, trading and building
Trade	Buy cargo in one port and sell it in another	Finance	Net worth, cash, the bank credit line, shareholdings and the ledger
Contracts	Freight offers from shippers	News	Headlines that report what happened in the simulation

The clock and the controls

- The game starts paused. Press Play at the top, or the space bar, to start and pause.
- One day lasts 12 seconds at Play speed. The menu lets you change this to 30, 20, 12, 6 or 2 seconds.
- The game counts 360 days in a year and closes a quarter every 90 days. It saves itself every 30 days, in the browser you are playing in. Esc closes a window.

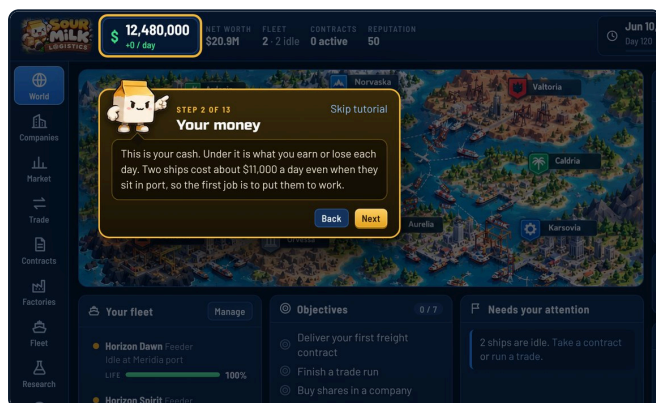
1.2 The guided tutorial, step by step

The studio's milk-carton mascot is your guide. A spotlight moves across the real game screen and walks you through one real turn of play in 13 steps. Five of the steps wait for you to do something; you can also skip a step or the whole tour. Four of the steps are pictured after the table.

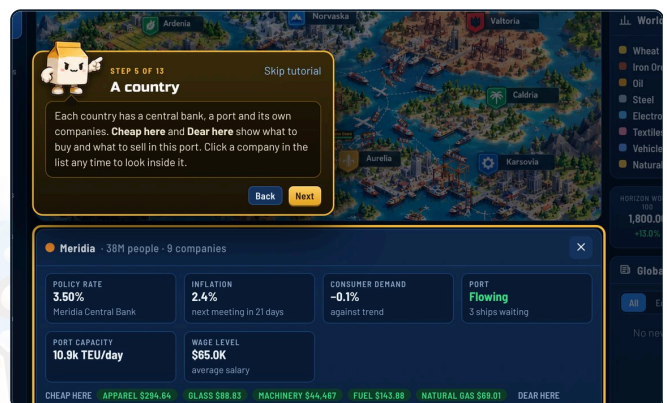
STEP	WHAT THE GUIDE SHOWS YOU	WHAT YOU DO	FINANCE IDEA
1 Welcome	You run a small shipping company with your starting cash and two feeder ships. A whole economy is already at work around you.	Read, then Next	The firm as a going concern
2 Your money	Your cash is at the top. Under it is what you earn or lose each day. Two ships cost about \$11,000 a day even when they sit in port, so the first job is to put them to work.	Read, then Next	Fixed costs and cash burn
3 The clock	The date is on the left and a gold bar shows the day passing. One day lasts 12 seconds.	Press the Play button	Time and daily accruals
4 The living world	Gold ships are yours. Every other ship is a real shipment from one company to another. Click one to see its cargo, route and risks.	Click any island	Supply chains
5 A country	Each country has a central bank, a port and its own companies. "Cheap here" and "Dear here" show what to buy and what to sell in this port.	Read, then Next	Price differences between markets
6 Earning: contracts	Shippers pay you to carry their cargo.	Open Contracts	Service revenue
7 Take your first contract	Each offer shows the cargo, the route, what it pays and the deadline. Late cargo pays less.	Press "Assign ship" on an offer, then pick a ship that arrives on time	Contract terms and penalties
8 Earning: trade	You can also buy cargo with your own money and sell it in another port. The table lists what your idle ship can buy here, the best place to sell it, and the profit after fuel. Exporters sell cheap; importers pay more.	Read, then Next	Arbitrage and working capital
9 Inside a company	You open Karsovia Motors, Factory 01. Materials come in, move down the line at the factory's real pace, and leave through the port. The slowest stage sets the pace for everything.	Click the Painting building (the red warning)	Capacity and bottlenecks
10 Find out why	When a company struggles, its Overview lists the problems with a "Why?" chain under each: a shortage leads to a late supplier, which leads to a congested port. Press Look to follow it.	Read, then Next	Root-cause analysis of results

STEP	WHAT THE GUIDE SHOWS YOU	WHAT YOU DO	FINANCE IDEA
11 Invest (Normal) Own it (Pro)	Normal: buy shares in any company for dividends and gains; other companies manage themselves. Pro: with 5% you vote on board proposals, with 20% you bring your own, and with 50% you run the company.	Read, then Next	Equity, dividends and control
12 Build your own	Later, found your own producer: pick a product, a country and a size. It needs time to build, then workers, materials and customers before it earns.	Read, then Next	Capital budgeting
13 You are ready	The World screen keeps a list of objectives and anything that needs your attention. The gear holds saving, the day length and this tutorial.	Press "Start playing"	Goals and performance measures

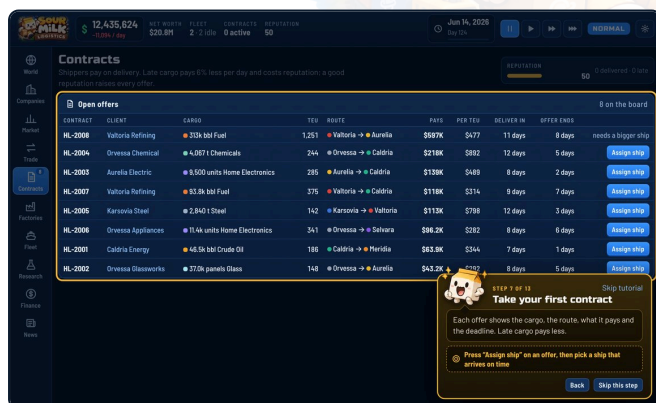
You can replay the tutorial at any time from the gear menu: Play the tutorial. "How it works" in the same menu summarises the rules of the simulation, the ownership tiers and the keys. Where it is available, Ask advisor at the top lets you ask the guide a question about your current game in your own words.



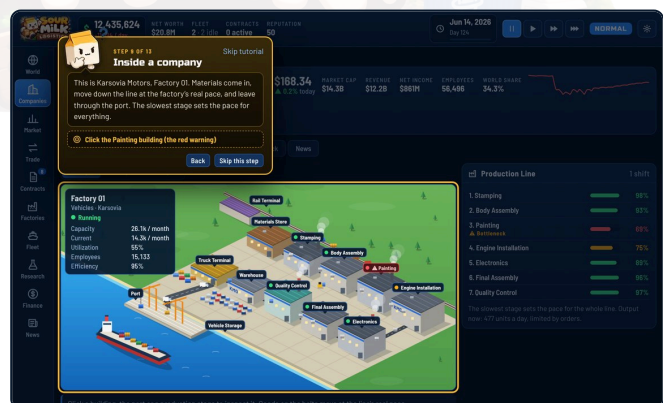
Step 2 · Your money. The spotlight is on your cash and daily result.



Step 5 · A country. Meridia's policy rate, inflation, port and wages.



Step 7 · Take your first contract. Eight offers are on the board.

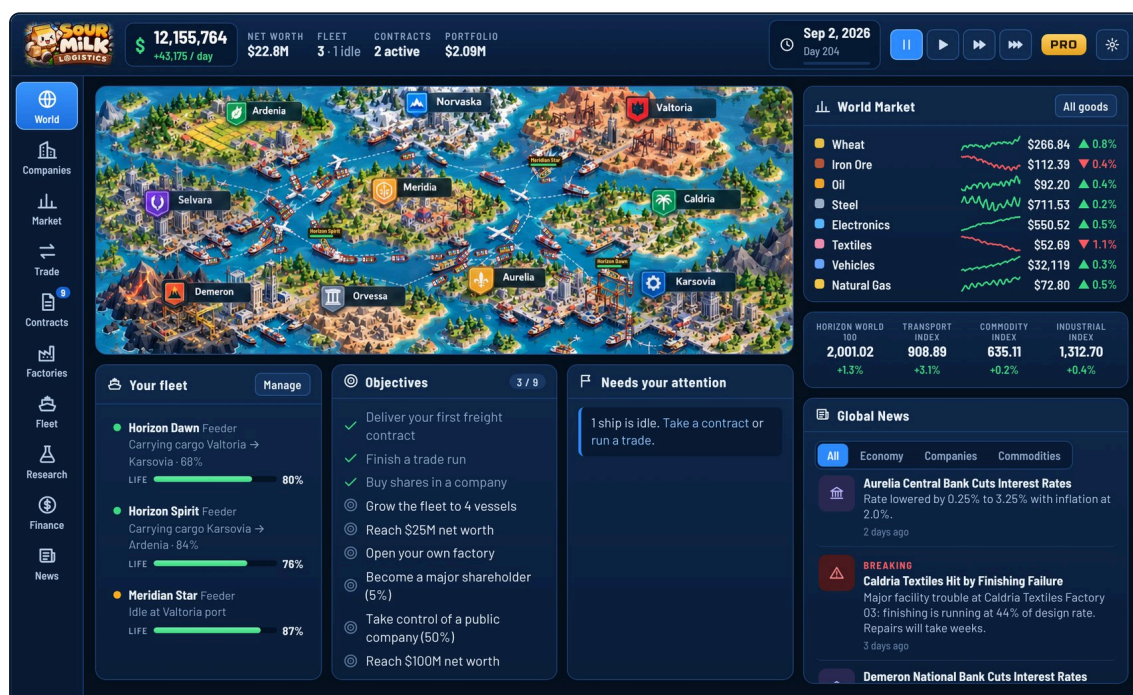


Step 9 · Inside a company. Painting is the bottleneck at Karsovia Motors.

1.3 Your objectives after the tutorial

The World screen lists nine objectives. They are a sensible order in which to learn the game, and each one is a finance milestone.

	OBJECTIVE	WHAT IT TEACHES	MODE
1	Deliver your first freight contract	Earning revenue on time	Both
2	Finish a trade run	Buying and selling on your own account	Both
3	Buy shares in a company	Becoming an investor	Both
4	Grow the fleet to 4 vessels	Reinvesting profit in capacity	Both
5	Reach \$25M net worth	Building equity	Both
6	Open your own factory	A long-term capital project	Both
7	Become a major shareholder (5%)	Shareholder voting rights	Pro
8	Take control of a public company (50%)	A takeover	Pro
9	Reach \$100M net worth	Sustained growth	Both



In the game. The World screen in a Pro-mode game on day 204: the live map, world market prices, your fleet, the news, and the objectives list with three of nine done.

Finance modules

Each module names the courses it supports, states the rule the game uses, explains the finance behind it, and works one example with the game's own numbers. Modules 8 and 9 need Pro mode.

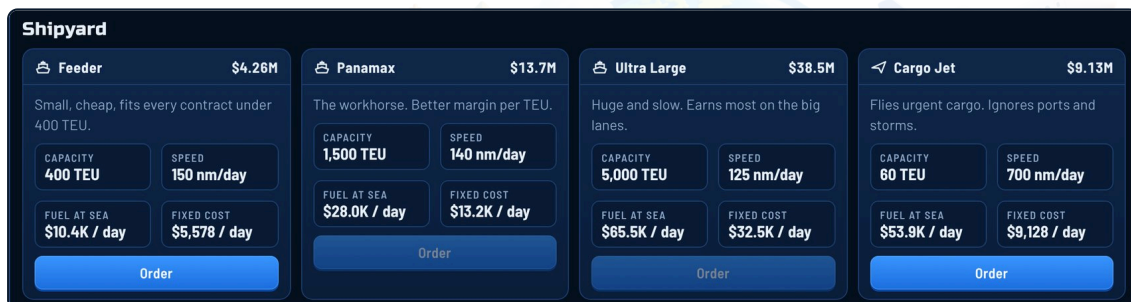
MODULE 1 Cost structure and break-even

Supports: Managerial Accounting · Introduction to Corporate Finance · Operations Management

IN THE GAME

Ships cost money every day, even in port, and burn fuel when they sail. There are four classes to buy at the shipyard on the Fleet screen.

SHIP CLASS	CAPACITY	PRICE	FIXED COST A DAY	PER TEU A DAY	BEST USE
Feeder	400 TEU	\$4.2M	\$5,500	\$13.75	Small and cheap; fits every contract under 400 TEU
Panamax	1,500 TEU	\$13.5M	\$13,000	\$8.67	The workhorse; better margin per TEU
Ultra Large	5,000 TEU	\$38.0M	\$32,000	\$6.40	Huge and slow; earns most on the big lanes
Cargo Jet	60 TEU	\$9.0M	\$9,000	\$150.00	Urgent cargo; ignores ports and storms



In the game. The shipyard on the Fleet screen. Prices and daily costs are a little above the starting values in the table because the game's price level has risen since day one.

THE FINANCE BEHIND IT

A **fixed cost** is paid whether or not you sell anything; a **variable cost**, such as fuel, rises with activity. A firm with high fixed costs has high **operating leverage**: profit rises fast once the fixed costs are covered and falls fast when capacity sits idle. The table also shows **economies of scale**: a bigger ship costs less per unit of capacity, but only if you can fill it.

Break-even: contribution per day = fixed cost per day

Contribution = revenue minus variable costs (fuel). Below this line the company loses cash every day.



Fixed cost per TEU of capacity per day. The largest ship costs less than half as much per unit as the smallest. The Cargo Jet (\$150.00) is left off the scale.

WORKED EXAMPLE

You start with two Feeders. Fixed cost = $2 \times \$5,500 = \$11,000$ a day, or \$330,000 over 30 days. The fleet must earn \$330,000 after fuel in a month before the company makes any profit. Over a 360-day year one Feeder costs \$1.98M to keep, which is 47% of its purchase price, so an idle ship destroys value quickly.

Try it. Play 30 days and note how many days each ship spent carrying cargo. Utilisation = days with cargo ÷ total ship-days. Compare your daily operating income on the Finance screen with the \$11,000 break-even line.

MODULE 2 Revenue, contracts and reputation

Supports: Financial Accounting · Supply Chain Management · Risk Management

IN THE GAME

- A contract pays when the cargo is delivered. Pay rises with the size of the cargo, the distance and the price of fuel.
- **Late cargo pays 6% less for each day late**, down to a floor of 40% of the agreed pay.
- **Reputation** runs from 0 to 100. An on-time delivery adds 1.5 points; a late one removes 3.
- New offers are priced by your reputation: $\text{pay} = \text{base pay} \times (1 + (\text{reputation} - 50) \div 400)$.
- About one offer in ten is urgent air cargo that needs a Cargo Jet and pays roughly five times the normal rate. Of the rest, about 58% fit a Feeder, 30% need a Panamax and 12% need an Ultra Large.
- An open offer expires after 4 to 8 days. In an online match the first team to take a contract gets it.

Contracts									
Shippers pay on delivery. Late cargo pays 6% less per day and costs reputation; a good reputation raises every offer.									
								REPUTATION	24 delivered · 0 late
								86	
In progress									
CONTRACT	CLIENT	CARGO	TEU	ROUTE	SHIP	STATUS	DUE	PAYS	
HL-2101	Karsovia Petrofuel	68.8k bbl Fuel	279	Karsovia → Ardenia	Horizon Spirit	Carrying cargo Karsovia → Ardenia - 84%	in 4 days	\$529K	
HL-2112	Valtoria Refining	65.5k bbl Fuel	262	Valtoria → Karsovia	Horizon Dawn	Carrying cargo Valtoria → Karsovia - 88%	in 8 days	\$243K	
Open offers									
CONTRACT	CLIENT	CARGO	TEU	ROUTE	PAYS	PER TEU	DELIVER IN	OFFER ENDS	
HL-2114	Caldria Fibers	14.3k t Cotton	1,425	Caldria → Ardenia	\$1.95M	\$1,370	13 days	2 days	needs a bigger ship
HL-2119	Caldria Harvest	18.3k t Wheat	730	Caldria → Aurelia	\$315K	\$432	10 days	2 days	needs a bigger ship
HL-2116	Ardenia Plantations	23.3k t Rubber	1,395	Ardenia → Selvara	\$270K	\$193	5 days	2 days	needs a bigger ship
HL-2118	Karsovia Motors	140 units Vehicles	279	Karsovia → Meridia	\$281K	\$934	11 days	2 days	Assign ship
HL-2120	Valtoria Refining	50.0k bbl Fuel	200	Valtoria → Selvara	\$253K	\$1,263	15 days	7 days	Assign ship
HL-2117	Norvaska Home	5,850 units Furniture	351	Norvaska → Demeron	\$250K	\$711	10 days	3 days	Assign ship
HL-2122	Norvaska Mining	6,033 t Iron Ore	181	Norvaska → Aurelia	\$193K	\$1,069	15 days	8 days	Assign ship
HL-2113 AIR	Norvaska Mining	2,000 t Iron Ore	60	Norvaska → Selvara	\$154K	\$2,562	-1 days	1 days	needs a jet
HL-2121 AIR	Norvaska Mining	1,100 t Iron Ore	33	Norvaska → Selvara	\$102K	\$3,091	3 days	6 days	needs a jet
Recently delivered									
HL-2108	Aurelia Pharma	20.4k cases Pharmaceuticals	204	Aurelia → Valtoria	\$115K	ON TIME	today		

In the game. The Contracts screen after 24 deliveries, none late: reputation has climbed to 86.

Offers too big for a Feeder are marked "needs a bigger ship" and urgent air cargo "needs a jet".

THE FINANCE BEHIND IT

Revenue is **recognised when the service is performed**, here at delivery, not when the contract is signed. A **penalty clause** turns operational delay into a financial loss you can measure. Reputation behaves like an **intangible asset**: it does not appear in net worth, but it raises every future cash flow. The asymmetry matters: one late delivery cancels two on-time ones.

WORKED EXAMPLE

A \$200,000 contract arrives 3 days late. Pay = $\$200,000 \times (1 - 3 \times 6\%) = \mathbf{\$164,000}$, a loss of \$36,000, and reputation falls by 3. At reputation 80, each new offer pays $1 + (80 - 50) \div 400 = \mathbf{7.5\% \text{ more}}$ than at reputation 50; at reputation 100 the premium is 12.5%.

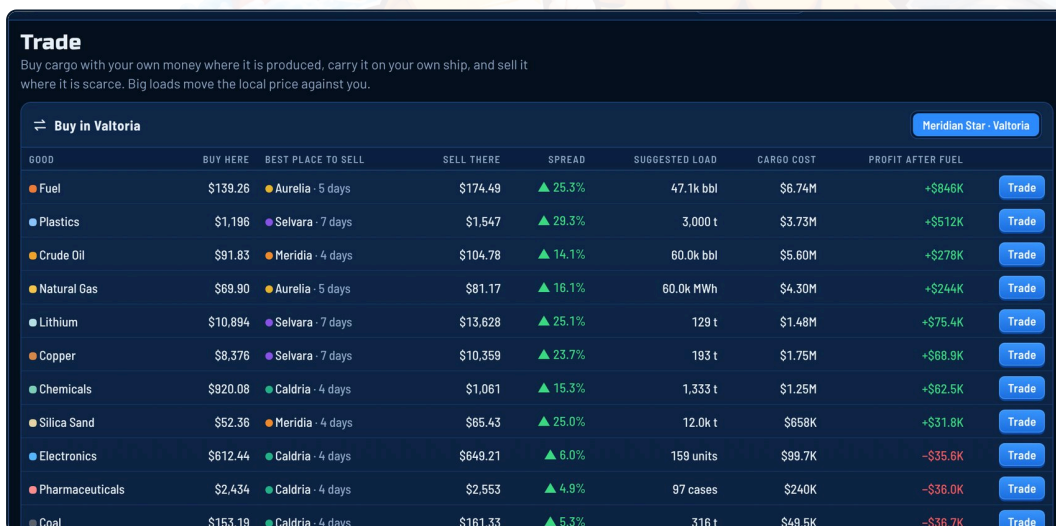
Try it. Before assigning a ship, compare the arrival estimate with the deadline. Decide the number of spare days you want as a safety margin when a storm or a congested port is on the route.

MODULE 3 Trade, arbitrage and working capital

Supports: Microeconomics · Investments · International Trade and Finance

IN THE GAME

- On the Trade screen an idle ship buys a good in its port and sells it in another. Ports that export a good sell it cheapest; ports that import it pay most.
- You buy at the local price $\times (1.01 + \text{slippage})$ and sell at the local price $\times (0.99 - \text{slippage})$.
- Slippage = $0.35 \times \text{your quantity} \div \text{the depth of that port's market}$. Big loads move the local price against you.
- The world trades 26 goods: 11 raw materials, 9 industrial goods and 6 consumer goods. The Market screen marks each one Short, Tight, Balanced or Glut, because prices move with real stock levels.
- The cargo is bought with your own cash, and the Trade screen shows what it is worth at the destination while it is still at sea.



GOOD	BUY HERE	BEST PLACE TO SELL	SELL THERE	SPREAD	SUGGESTED LOAD	CARGO COST	PROFIT AFTER FUEL	
Fuel	\$139.26	Aurelia - 5 days	\$174.49	▲ 25.3%	47.1k bbl	\$6.74M	+\$846K	Trade
Plastics	\$1,196	Selvara - 7 days	\$1,547	▲ 29.3%	3,000 t	\$3.73M	+\$512K	Trade
Crude Oil	\$91.83	Meridia - 4 days	\$104.78	▲ 14.1%	60.0k bbl	\$5.60M	+\$278K	Trade
Natural Gas	\$69.90	Aurelia - 5 days	\$81.17	▲ 16.1%	60.0k MWh	\$4.30M	+\$244K	Trade
Lithium	\$10.894	Selvara - 7 days	\$13.628	▲ 25.1%	129 t	\$1.48M	+\$75.4K	Trade
Copper	\$8,376	Selvara - 7 days	\$10,359	▲ 23.7%	193 t	\$1.75M	+\$68.9K	Trade
Chemicals	\$920.08	Caldria - 4 days	\$1,061	▲ 15.3%	1,333 t	\$1.25M	+\$62.5K	Trade
Silica Sand	\$52.36	Meridia - 4 days	\$65.43	▲ 25.0%	12.0k t	\$858K	+\$31.8K	Trade
Electronics	\$612.44	Caldria - 4 days	\$649.21	▲ 6.0%	159 units	\$99.7K	-\$35.6K	Trade
Pharmaceuticals	\$2,434	Caldria - 4 days	\$2,553	▲ 4.9%	97 cases	\$240K	-\$36.0K	Trade
Coal	\$153.19	Caldria - 4 days	\$161.33	▲ 5.3%	316 t	\$49.5K	-\$36.7K	Trade

In the game. The Trade screen for a ship idle in Valtoria. Each row shows the buy price, the best port to sell in, the spread, a suggested load, the cargo cost and the profit after fuel.

THE FINANCE BEHIND IT

Arbitrage is buying where a good is cheap and selling where it is dear. In theory the **law of one price** removes the gap; in practice transport cost, time and risk keep it open, and that gap is your margin. The 1% on each side is a **bid-ask spread**. Slippage is **price impact**, the cost of trading in a market that is not perfectly liquid. Cash tied up in cargo is **working capital**: it earns nothing until the ship arrives, and the price can move while you wait.

Trade profit = quantity × (sell price – buy price) – fuel

Return on the voyage = trade profit ÷ cash paid for the cargo. Divide by the days at sea to compare routes.

WORKED EXAMPLE

Plastics cost \$1,350 a tonne in an exporting port and \$1,600 in an importing port. With a small load and no slippage you buy at $\$1,350 \times 1.01 = \$1,363.50$ and sell at $\$1,600 \times 0.99 = \$1,584.00$, a gross margin of **16.2%** before fuel. With a load big enough to cause 2% slippage on each side you buy at \$1,390.50 and sell at \$1,552.00, and the margin falls to **11.6%**.

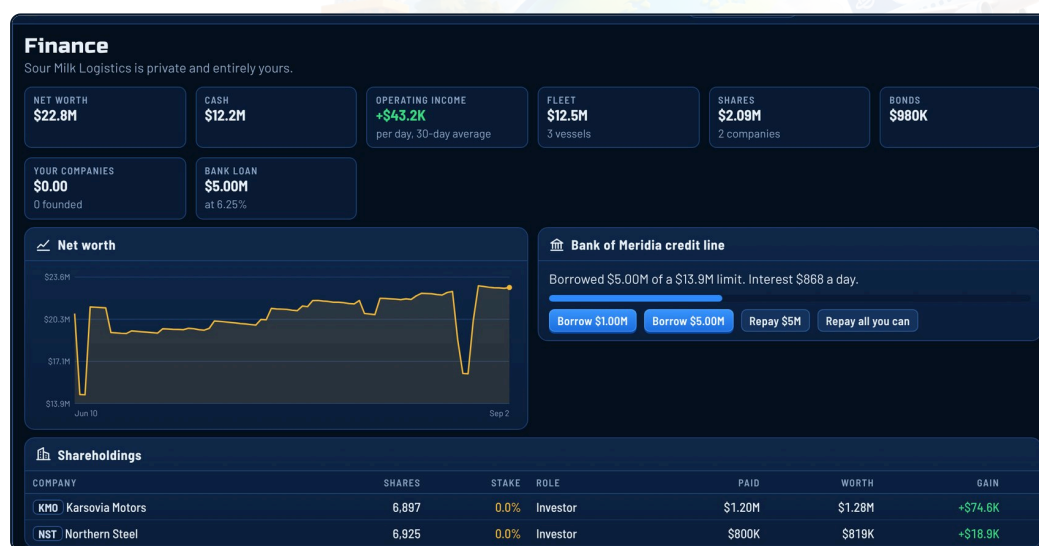
Try it. Run the same route twice, once with a full load and once with half. Compare profit per unit. Halving the quantity halves the slippage.

MODULE 4 Borrowing, interest and leverage

Supports: Corporate Finance · Financial Institutions Management · Money and Banking

IN THE GAME

- The Finance screen has a credit line from the Bank of Meridia. You borrow in steps of \$1M, \$5M, \$25M or \$100M and repay at any time.
- Limit:** the bank lends up to half of your total assets, counted as your net worth plus the loan you already have.
- Rate:** Meridia's central bank rate plus 2.5 percentage points. Interest is charged every day on a 360-day year.
- The rate floats. When Meridia's central bank moves its rate, your interest cost moves with it.



In the game. The Finance screen: \$5.00M borrowed of a \$13.9M limit at 6.25%, costing \$868 a day. Meridia's rate has risen to 3.75%, so the loan rate has risen with it.

THE FINANCE BEHIND IT

Debt lets a company own more assets than its equity alone would buy. That is **financial leverage**. It raises the return on equity when the assets earn more than the loan costs and lowers it when they earn less. A floating rate adds **interest-rate risk**: the cost of the loan is set by a central bank you do not control.

Daily interest = loan × (policy rate + 2.5%) ÷ 360

ROE = ROA + (ROA – interest rate) × Debt ÷ Equity

ROA is the return the assets earn; ROE is the return to the owner after interest. Tax is ignored.

WORKED EXAMPLE

Meridia's rate starts at 3.50%, so the loan costs 6.00%. On \$5M: $\$5,000,000 \times 6\% \div 360 = \text{\$833 a day}$, or \$300,000 a year. If the central bank raises its rate by 0.50 points the cost becomes \$903 a day.

Borrow to the limit (debt equal to equity) and buy ships that earn 12% on their cost: $\text{ROE} = 12\% + (12\% - 6\%) \times 1 = \textbf{18\%}$. If the same ships earn only 3%: $\text{ROE} = 3\% + (3\% - 6\%) \times 1 = \textbf{0\%}$. Leverage doubled the gap in both directions.

Try it. Before borrowing for a ship, estimate what it will earn per day after fuel and fixed cost, turn that into a yearly percentage of its price, and compare it with the loan rate shown on the Finance screen.

MODULE 5 Capital budgeting

Supports: Corporate Finance · Project Appraisal · Strategic Management

IN THE GAME

You can commit cash to three kinds of long-term project: ships (Module 1), research and your own factory. Every decision shows its benefit, cost, risk and time before you commit.

RESEARCH PROJECT	COST	TIME	BENEFIT
Efficient Hulls	\$2.5M	60 days	Fleet fuel use -15%
Fast Turnaround	\$3.0M	75 days	Loading and unloading take 40% less time
Weather Routing	\$4.0M	90 days	Ships sail 10% faster and lose half as much time in storms
Key Account Desk	\$3.5M	80 days	Freight contracts pay 8% more
Trade Analytics	\$2.0M	50 days	Shows the best buy-sell spreads between ports; trading slippage -30%
Lean Construction	\$6.0M	120 days	New factories cost 12% less and finish 20% faster

Research
One project at a time. Each gives Sour Milk Logistics a permanent edge.

Efficient Hulls \$2.50M Fleet fuel use -15%. 60 days Start	Fast Turnaround \$3.00M Loading and unloading take 40% less time. 75 days Start	Weather Routing \$4.00M Ships sail 10% faster and lose half as much time in storms. 90 days Start	Key Account Desk \$3.50M Freight contracts pay 8% more. 80 days Start
Trade Analytics DONE Shows the best buy-sell spreads between ports. Trading slippage -30%.	Lean Construction \$6.00M New factories cost 12% less and finish 20% faster. 120 days Start		

In the game. The Research screen. Trade Analytics is done; the other five projects are waiting.

Only one research project runs at a time. **A factory** is the largest project: you pick a product, a country and one of three sizes, sized at about 0.25%, 1% or 4% of world demand. It takes roughly 150, 240 or 330 days to build. The cost has two parts: the plant, which scales with expected yearly revenue and how capital-intensive the product is, and working capital of about 14% of yearly revenue. A new factory earns nothing until it is built, staffed, supplied and its output is sold.

Factories

Find your own producer anywhere in the world. It needs time to build, then workers, materials and customers before it earns anything.

You do not run any company yet. Found one below, or buy 50% of a listed company on the Companies screen.

Found a new company

Product

Country

Size

Furniture

Caldria

Workshop

Plant

Complex

TOTAL COST

\$9.61M

\$7.49M plant + \$2.12M working capital

OPENS

Jan 30, 2027

150 days of construction

CAPACITY

2,924 units / month

0.25% of world demand

SALES IF IT ALL SELLS

\$15.1M / year

materials are 52% of the price

Materials it will buy: Timber from Ardenia Forestry (7 days), Textiles from Caldria Textiles (2 days), Steel from Karsovia Steel (4 days).

Caldria buys about \$530M of furniture a year from 1 supplier. Wages here are 55% of the world average. Your share starts at zero and builds as you deliver.

You have \$12.2M.

Found it for \$9.61M

In the game. Founding a Workshop-size furniture factory in Caldria: \$9.61M in total, of which \$2.12M is working capital (14% of the \$15.1M yearly sales), and 150 days to build.

THE FINANCE BEHIND IT

Capital budgeting asks whether the future cash a project brings in is worth more than the cash it costs today.

Payback is the time to recover the outlay. **Net present value (NPV)** discounts each future cash flow at your cost of capital. Research projects are **mutually exclusive in time**, so you rank them. A factory shows the **J-curve**: cash goes out for months before any comes back.

$$\text{NPV} = -C_0 + CF_1 \div (1 + r) + CF_2 \div (1 + r)^2 + \dots + CF_n \div (1 + r)^n$$

$$\text{Payback} = \text{initial cost} \div \text{extra cash per period}$$

Accept a project when NPV is positive. A natural discount rate in the game is your loan rate.

WORKED EXAMPLE

Key Account Desk costs \$3.5M and adds 8% to freight pay. Break-even freight revenue = $\$3,500,000 \div 0.08 = \43.75M . If your fleet earns \$60,000 a day from contracts (an assumed figure), the project adds \$4,800 a day and pays back in about 729 days, a little over two game years, after the 80 days of research. A small fleet should wait; a large one should not.

Try it. Rank the six research projects for your own company by payback, using your last 30 days of fuel cost, freight income and trade volume.

MODULE 6 Reading a company

Supports: Financial Statement Analysis · Intermediate Accounting · Operations Management

IN THE GAME

- Any of the 79 companies can be opened. Its tabs are Overview, Operations, Production, Supply Chain, Facilities, Logistics, Employees, Financials, Stock, Bonds, News and Management.
- Nothing is faked. A company's profit comes from what it actually sold. To sell it must hold stock; to hold stock it must produce; to produce it needs every material on site, working lines and staff.
- The Overview lists current problems, each with a "Why?" chain you can follow to its cause, such as a late supplier or a congested port.
- Companies close their books every quarter and pay 22% tax on pre-tax profit.

World · Karsovia · Karsovia Motors · Financials

KARSOVIA MOTORS (KMO) **\$184.89** MARKET CAP **\$15.7B** REVENUE **\$12.5B** NET INCOME **\$1.10B** EMPLOYEES **54,318** WORLD SHARE **33.3%**
 Vehicles · Karsovia · CEO Adrian Varek
 INVESTOR · 0.01%
 Buy shares Sell

Overview Operations Production Supply Chain Facilities Logistics Employees **Financials** Stock Bonds News Management

Income statement every line comes from simulated operations

	Q2 2028	Q3 2028	THIS QUARTER SO FAR
Revenue	\$3.05B	\$3.07B	\$830M
Materials	(\$1.22B)	(\$1.22B)	(\$330M)
Labor	(\$713M)	(\$690M)	(\$181M)
Energy	(\$99.0M)	(\$95.4M)	(\$25.4M)
Maintenance	(\$47.0M)	(\$46.9M)	(\$12.5M)
Outbound freight	(\$46.4M)	(\$46.9M)	(\$12.9M)
Selling & admin	(\$212M)	(\$213M)	(\$57.2M)
R&D	(\$126M)	(\$127M)	(\$33.9M)
Storage	(\$5.87M)	(\$5.79M)	(\$1.55M)
Depreciation	(\$104M)	(\$104M)	(\$27.8M)
One-offs	\$0.00	\$0.00	\$0.00
Operating profit	\$477M	\$522M	\$147M
Interest	(\$32.1M)	(\$32.1M)	(\$8.57M)
Tax	(\$98.0M)	(\$108M)	(\$30.5M)
Net income	\$347M	\$382M	\$108M
Earnings per share	\$4.08	\$4.50	\$1.27

In the game. The Financials tab of Karsovia Motors: an income statement by quarter, from revenue down to earnings per share. Tax is 22% of profit after interest.

THE FINANCE BEHIND IT

Financial statements are the result of operations, not the cause. **Ratio analysis** tells you that something is wrong; the operating data tells you why. In the game you can check both, which is rare for a student analyst.

RATIO	FORMULA	WHAT IT TELLS YOU
Net margin	Net income ÷ revenue	How much of each sales dollar is kept as profit
Net debt to operating cash flow	(Bank debt + bonds – cash) ÷ operating cash flow	How many years of cash from operations it would take to repay debt
Interest cover	Operating earnings ÷ interest bill	How easily the company pays its lenders
Capacity utilisation	Demand ÷ factory capacity	Whether fixed assets are being used; the game flags low demand below 60%
Price to earnings (P/E)	Share price ÷ earnings per share	What investors pay for each dollar of profit
Dividend yield	Dividend per share ÷ share price	The cash return from holding the share

Try it. Pick one company that is losing money. Write five lines: its net margin, its interest cover, its utilisation, the first problem on its Overview, and the root cause at the end of the Why chain.

MODULE 7 Shares, dividends and control

Supports: Investments · Corporate Governance · Mergers and Acquisitions

IN THE GAME

- You can buy shares in any listed company. Dividends from holdings arrive every quarter.
- A share price drifts toward a fair value based on the company's results, pushed by market sentiment and daily noise.
- Your own orders move the price: impact = $10\% \times \sqrt{(\text{shares traded} \div \text{shares outstanding})}$. You pay about half of that impact on average.
- In Pro mode your stake decides your power:

Under 5%

Investor

Dividends only.

5% +

Major holder

Votes on board proposals.

20% +

Strategic

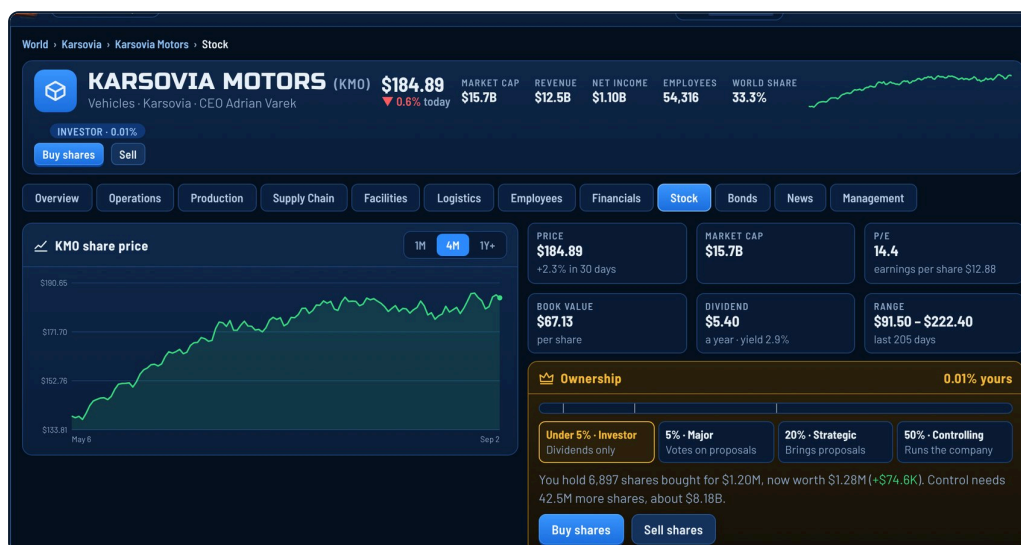
Brings proposals to the board.

50% +

Controlling

Decides directly.

- With control you set the company's finance: issue shares, buy back shares, issue bonds, repay debt, raise or cut the dividend, acquire another company, invest in R&D, or replace the CEO.
- An acquisition costs the target's market value plus a 30% premium and closes in 60 days.
- A company that runs short of cash issues new shares at a discount, diluting existing holders by 11%. In a restructuring, creditors take a 50% haircut and receive new shares, and the old shareholders keep one third of the company.



In the game. The Stock tab: price history, P/E, book value, dividend yield and the ownership ladder. Here control of Karsovia Motors would take 42.5M more shares, about \$8.18B.

THE FINANCE BEHIND IT

Price is not value. The market price can sit above or below fair value, and an investor earns from the gap closing as well as from **dividends**. **Dividend policy** is a trade-off: paying out rewards shareholders now but leaves less cash for investment and debt repayment. Buyers pay a **control premium** because control lets them change how a company is run. **Dilution** and the restructuring rule show the **priority of claims**: creditors are paid before shareholders.

WORKED EXAMPLE

You buy 4% of a company in one order. $\text{Impact} = 10\% \times \sqrt{0.04} = 2.0\%$: the price ends 2% higher and your average cost is about 1% above the starting price. Buying 25% in one order moves the price 5%. To acquire a company with a market value of \$400M, the offer is $\$400\text{M} \times 1.30 = \520M .

Try it. Pro mode: reach 5% of one company and read the board proposals you can now vote on. For each, write down who gains: shareholders, creditors or management.

MODULE 8 · PRO Bonds and interest-rate risk

Supports: Fixed Income Securities · Financial Institutions Management · Risk Management

IN THE GAME

- The Market screen has a Bonds tab. Each of the ten countries has a **government bond**, priced as a 10-year bond. Its coupon is the country's starting policy rate plus 0.8 points; its yield is the current policy rate plus 0.8 points. Coupons are paid every quarter.

- A **corporate bond** yields the country's policy rate, plus a credit spread for the company's rating, plus a term premium of 0.12 points for each year left to maturity (up to 8 years).
- Buying or selling a bond costs 0.2% of the price.

CREDIT RATING	AAA	AA	A	BBB	BB	B	CCC
Credit spread (percentage points)	0.5	0.8	1.2	1.9	3.2	5.0	9.0

Market						Commodities Stocks Bonds Central banks					
Government bonds, 10 years						Corporate bonds, by yield					
COUNTRY	COUPON	YIELD	PRICE	YOU HOLD		ISSUER	RATING	COUPON	MATURES	YIELD	PRICE
Ardenia	5.05%	5.05%	100.0	-	Trade	NVM Norvaska Mining	CCC	5.23%	Dec 31, 2030	12.53%	78.4
Norvaska	3.80%	3.80%	100.0	-	Trade	NVM Norvaska Mining	CCC	4.22%	Oct 16, 2027	12.14%	92.0
Valtoria	5.80%	5.80%	100.0	-	Trade	CFB Caldria Fibers	B	7.69%	May 31, 2030	11.21%	89.6
Selvara	4.05%	3.80%	102.0	-	Trade	CFB Caldria Fibers	B	8.03%	Mar 25, 2030	11.18%	91.0
Meridia	4.30%	4.55%	98.0	\$1.00M	Trade	CFB Caldria Fibers	B	7.36%	Sep 20, 2027	10.88%	96.6
Caldria	6.80%	6.55%	101.8	-	Trade	DMD Demeron Ore	BBB	8.83%	Nov 24, 2033	8.78%	100.3
Demeron	7.30%	6.80%	103.5	-	Trade	DMD Demeron Ore	BBB	8.90%	Nov 19, 2030	8.41%	101.7
Orvessa	4.80%	5.05%	98.1	-	Trade	DMD Demeron Ore	BBB	8.48%	Mar 22, 2029	8.21%	100.6
Aurelia	4.55%	4.05%	104.0	-	Trade	DCL Demeron Coal	A	8.04%	Jan 1, 2037	8.16%	99.2
Karsovia	5.30%	5.30%	100.0	-	Trade	CEN Caldria Energy	A	8.36%	Jan 31, 2035	7.91%	102.7
						DCL Demeron Coal	A	7.62%	Oct 17, 2031	7.82%	99.2

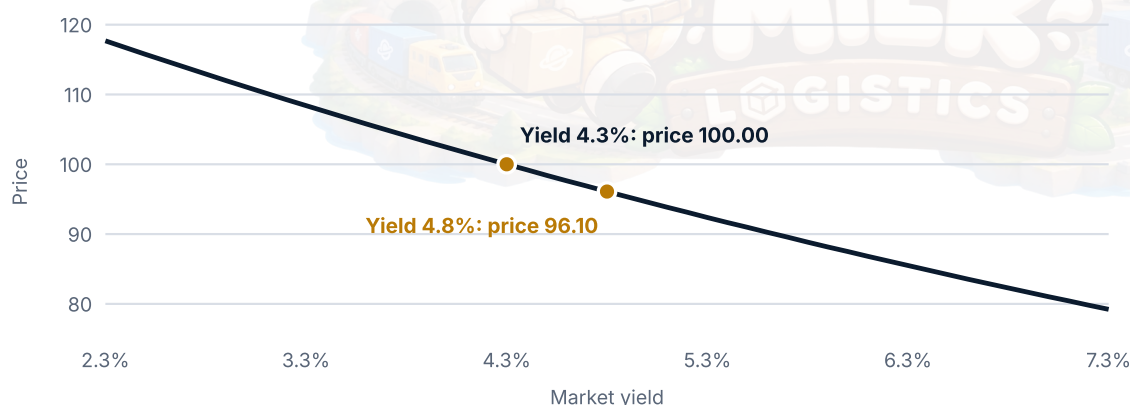
In the game. The Bonds tab. After a 0.25-point rate rise, Meridia's 4.30% government bond yields 4.55% and trades at 98.0, which is what the formula below gives. CCC-rated corporate bonds yield over 12%.

THE FINANCE BEHIND IT

A bond's price is the **present value** of its coupons and its face value, discounted at the market yield. When yields rise, prices fall, and the longer the bond the larger the fall. **Duration** measures that sensitivity. A **credit spread** is the extra yield investors demand for the risk that the issuer does not pay.

$$\text{Price} = 100 \times [(c \div y) \times (1 - (1 + y)^{-n}) + (1 + y)^{-n}]$$

c = coupon rate, y = yield, n = years to maturity. Price change $\approx$ -modified duration $\times$ change in yield.



Price of a 10-year bond with a 4.3% coupon at different market yields. The curve falls as yields rise, and it is steeper at low yields.

WORKED EXAMPLE

Meridia's government bond has a 4.3% coupon ($3.5\% + 0.8$). At a yield of 4.3% it is priced at 100.00. If the central bank raises its rate by 0.50 points the yield becomes 4.8% and the price falls to **96.10**, a loss of 3.9%. The bond's modified duration is about 8.0, and $-8.0 \times 0.50\% = -4.0\%$ is a close estimate.

A BBB company in Valtoria (policy rate 5.0%) has a bond with 5 years left and a 6% coupon. Yield = $5.0 + 1.9 + (0.12 \times 5) = 7.5\%$, so the price is **93.93**.

Try it. Buy one government bond and note its price. After the next rate decision in that country, compare the price change with your duration estimate.

MODULE 9 · PRO Central banks and inflation

Supports: Monetary Economics · Macroeconomics · Financial Institutions Management

IN THE GAME

Each country has its own central bank and policy rate. The Market screen's Central banks tab tracks them, and every rate decision appears in the News.

COUNTRY	CENTRAL BANK	STARTING RATE	COUNTRY	CENTRAL BANK	STARTING RATE
Ardenia	Bank of Ardenia	4.25%	Caldria	Bank of Caldria	6.00%
Norvaska	Norvaska Riksbank	3.00%	Demeron	Demeron National Bank	6.50%
Valtoria	Valtoria Reserve	5.00%	Orvessa	Bank of Orvessa	4.00%
Selvara	Selvara Monetary Authority	3.25%	Aurelia	Aurelia Central Bank	3.75%
Meridia	Meridia Central Bank	3.50%	Karsovia	Karsovia State Bank	4.50%

- Each central bank meets every 45 days and compares its rate with a target that depends on inflation.
- It moves by 0.25 points when it is more than 0.5 points from the target, by 0.50 points when it is more than 2 points away, and otherwise holds.
- Higher rates cool consumer demand, lift every company's interest bill, lower share valuations and push bond prices down.

Market						
Prices move with real stock levels. When producers cannot keep up, cover falls and the price climbs.						
COUNTRY	CENTRAL BANK	POLICY RATE	RATE, 2 YEARS	INFLATION	CONSUMER DEMAND	NEXT MEETING
Ardenia	Bank of Ardenia	4.25%		2.5%	+2.7%	11 days
Norvaska	Norvaska Riksbank	3.00%		2.4%	+1.4%	15 days
Valtoria	Valtoria Reserve	5.00%		2.4%	+0.5%	19 days
Selvara	Selvara Monetary Authority	3.00%		1.9%	-2.2%	23 days
Meridia	Meridia Central Bank	3.75%		2.8%	-0.1%	27 days
Caldria	Bank of Caldria	5.75%		1.9%	-0.2%	31 days
Demeron	Demeron National Bank	6.00%		1.6%	-0.9%	35 days
Orvessa	Bank of Orvessa	4.25%		2.7%	-0.6%	39 days
Aurelia	Aurelia Central Bank	3.25%		2.0%	-0.3%	43 days
Karsovia	Karsovia State Bank	4.50%		2.4%	+0.6%	2 days

In the game. The Central banks tab: each policy rate and its two-year path, inflation, consumer demand against trend, and the days to the next meeting.

THE FINANCE BEHIND IT

The game's central banks follow a **Taylor-type rule** with an inflation target of 2.5%: when inflation is above target they raise the rate by more than the rise in inflation, so the real rate goes up. The effects reach you through three channels: your **loan cost** (Module 4), **bond prices** (Module 8) and **customer demand**.

Target rate = (starting rate – 2.5) + inflation + 0.5 × (inflation – 2.5)

All in percent, limited to between 0.25% and 14%. At 2.5% inflation the target equals the starting rate.

WORKED EXAMPLE

Caldria starts at 6.00%. If its inflation runs at 5.0%, the target is $(6.0 - 2.5) + 5.0 + 0.5 \times (5.0 - 2.5) = 9.75\%$. The gap is 3.75 points, so the Bank of Caldria raises by 0.50 at its next meeting and keeps raising every 45 days while the gap stays wide. Bonds in Caldria lose value and its companies see weaker demand.

Try it. Follow one country for 180 days. Record inflation and the policy rate at each meeting and check each decision against the rule.

MODULE 10 Net worth, performance and risk

Supports: Corporate Finance · Risk Management · Financial Reporting

IN THE GAME

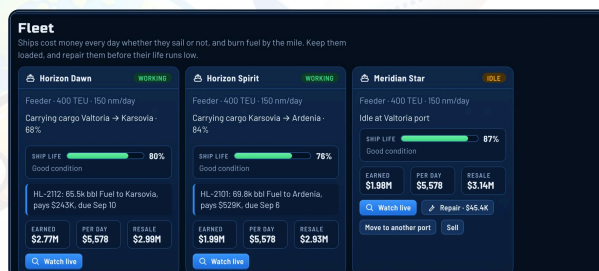
The Finance screen shows net worth and its history, cash, the 30-day average of operating income per day, the value of the fleet, shares, bonds and the companies you founded, the bank loan, and a ledger of every transaction. Net worth is also the default score in an online match.

Net worth = cash + ships + shares + bonds + companies you founded – bank loan

Shares and bonds are counted at today's market price, so net worth moves with the market. Cargo bought for a trade is not counted until it is sold, so net worth dips during a trade voyage.

The main operating risk is the condition of your ships:

- Every ship has a life bar from 0 to 100%. It wears down at sea and fastest in storms.
- Under 50% the ship sails slower. Under 25% it can break down and lose days.
- A repair is only possible while the ship is idle in port. Cost = share of life missing × ship price × 8%. It takes one day in dry dock for each 25% of life restored, rounded up (1 to 4 days).
- Storms and congested ports delay voyages. A Cargo Jet ignores both.



In the game. Ship cards on the Fleet screen: the life bar, what each ship has earned, its daily cost and resale value. The idle ship at 87% life can be repaired for \$45.4K.

THE FINANCE BEHIND IT

Net worth is the owner's **equity** at market value: assets minus debt. Its growth rate is the truest single measure of how well the company is run. **Risk management** means paying a known, small cost to avoid an uncertain, large one. A repair is such a cost: it is cheap compared with a breakdown that makes a contract late, cuts its pay by 6% a day and costs 3 reputation points.

WORKED EXAMPLE

A Feeder at 60% life: $40\% \times \$4.2M \times 8\% = \$134,400$ and 2 days in dry dock. A Panamax at 40% life: $60\% \times \$13.5M \times 8\% = \$648,000$ and 3 days. Compare that with the contracts the ship could not carry safely in its current state.

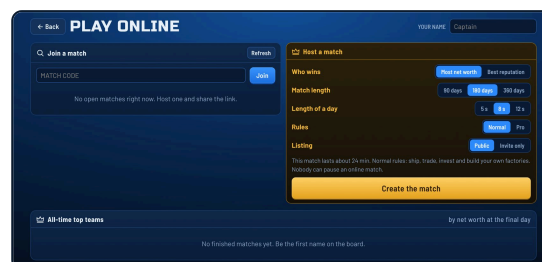
Try it. At day 90 and day 180, write down each line of the net worth formula from the Finance screen. Work out which line grew most and what decision caused it.

Online matches for a class

"Play online" on the title screen puts every team into one shared world. Every team faces the same prices and the same opportunities, so results can be compared.

How a match works

- Play solo or in a team of up to four. A team picks a name and gets a team code that lets a friend join at any time.
- Everyone shares one world: the same prices, the same contract board and the same companies.
- The first team to take a contract gets it. Big trades move the local price for all teams for a few weeks.
- Nobody can pause an online match. A leaderboard shows the standings and the time left.
- If you leave, your team keeps its company. Return with the match code and your team code.



In the game. The Play online lobby: join with a match code, or host a match and set its rules.

What the host sets

SETTING	OPTIONS	NOTE
Who wins	Most net worth · Best reputation	Under Best reputation, net worth breaks ties.
Match length	90, 180 or 360 game days	The standard options.
Length of a day	5, 8 or 12 seconds	Slower days leave more time to discuss.
Rules	Normal · Pro	Pro lets teams take over listed companies.
Listing	Public · Invite only	Invite only suits a class; share the link.

How long a match takes

MATCH LENGTH	5 S DAYS	8 S DAYS	12 S DAYS
90 days	7.5 min	12 min	18 min
180 days	15 min	24 min	36 min
360 days	30 min	48 min	72 min

Suggested team roles

ROLE	WATCHES
CEO	Objectives and final decisions
CFO	Cash, loan and net worth
Fleet manager	Contracts, deadlines, ship life
Analyst	Prices and companies

A suggested 90-minute class session

	STAGE	TIME	WHAT HAPPENS
1	Learn	20 minutes	Each student plays the guided tutorial solo and delivers one contract.
2	Team up	10 minutes	Teams agree roles and join the host's invite-only match.
3	Compete	36 minutes	A 180-day match with 12-second days; most net worth wins.
4	Debrief	24 minutes	Each team explains its result with the net worth formula.

Course map

Use this table to match the modules to a syllabus.

UNIVERSITY COURSE	MODULES	MAIN TOPICS PRACTISED
Managerial and Cost Accounting	1, 5, 6	Fixed and variable cost, break-even, capacity utilisation
Financial Accounting and Statement Analysis	2, 6, 10	Revenue recognition, ratios, equity
Corporate Finance	4, 5, 7, 10	Leverage, NPV and payback, dividend policy, valuation
Investments and Portfolio Management	3, 7, 8	Arbitrage, price impact, shares and bonds
Fixed Income Securities	8	Bond pricing, duration, credit spreads
Financial Institutions and Risk Management	4, 8, 9, 10	Floating-rate lending, interest-rate risk, operational risk
Monetary Economics and Macroeconomics	9	Policy rules, inflation targeting, transmission
Mergers, Acquisitions and Governance	7	Control tiers, takeover premium, dilution, restructuring
Operations and Supply Chain Management	1, 2, 6	Fleet utilisation, service reliability, bottlenecks
International Trade and Finance	3, 9	Price gaps between countries, national interest rates

Practice questions

All twelve use the game's rules as stated in Part 2. Answers follow.

- 1 Your two Feeders sit idle in port for 10 days. How much fixed cost do you pay?
- 2 A contract worth \$180,000 is delivered 4 days late. What does it pay, and what happens to reputation?
- 3 After how many days late does a contract reach its minimum pay?
- 4 What is the fixed cost per TEU of capacity per day for a Panamax, and how does it compare with a Feeder?
- 5 Your net worth is \$30M and you have already borrowed \$10M. How much more will the bank lend?
- 6 You borrow \$25M while Meridia's policy rate is 3.50%. What is the daily interest? What is it after a rise to 4.00%?
- 7 How much freight revenue must you earn for the Key Account Desk research to pay for itself?
- 8 A good costs \$1,350 in the buying port and \$1,600 in the selling port. With 2% slippage on each side, what is the margin per unit?
- 9 You buy 4% of a company's shares in one order. By how much does the price move?
- 10 Caldria's government bond has a 6.8% coupon. Its policy rate rises from 6.00% to 6.50%. What is the new price?
- 11 Caldria's inflation is 5.0%. What is the central bank's target rate, and what does it do at its next meeting?
- 12 A Panamax is at 40% life. What does the repair cost and how long does it take?

Answers

- 1 $2 \times \$5,500 \times 10 = \text{\$110,000}$.
- 2 $\$180,000 \times (1 - 4 \times 6\%) = \text{\$136,800}$. Reputation falls by 3 points.
- 3 The floor is 40% of pay, so $1 - 0.06 \times d = 0.40$ gives $d = \text{10 days}$.
- 4 $\$13,000 \div 1,500 = \text{\$8.67}$, against \$13.75 for a Feeder: 37% lower, if the ship is full.
- 5 The limit is $50\% \times (\$30\text{M} + \$10\text{M}) = \$20\text{M}$ in total, so **\\$10M more**.
- 6 $\$25\text{M} \times 6.00\% \div 360 = \text{\$4,167 a day}$. At 6.50%: **\\$4,514 a day**.
- 7 $\$3,500,000 \div 0.08 = \text{\$43.75M}$ of freight revenue.
- 8 Buy at $\$1,350 \times 1.03 = \$1,390.50$; sell at $\$1,600 \times 0.97 = \$1,552.00$. Margin = **\\$161.50 a unit, or 11.6%**, before fuel.
- 9 $10\% \times \sqrt{0.04} = \text{2.0\%}$. Your average cost is about 1% above the starting price.
- 10 The yield rises from 6.8% to 7.3%. Price = $100 \times [(0.068 \div 0.073) \times (1 - 1.073^{-10}) + 1.073^{-10}] = \text{96.54}$.
- 11 $(6.0 - 2.5) + 5.0 + 0.5 \times (5.0 - 2.5) = \text{9.75\%}$. The gap is more than 2 points, so it **raises by 0.50**.
- 12 $60\% \times \$13.5\text{M} \times 8\% = \text{\$648,000}$, and **3 days** in dry dock.

Glossary

TEU	Twenty-foot equivalent unit, the standard measure of container cargo and ship capacity.
Policy rate	The interest rate a central bank sets for its country.
Slippage	The amount a price moves against you because of the size of your own order.
Coupon	The interest a bond pays each year, as a percentage of its face value.
Yield	The return the market demands on a bond today; it sets the bond's price.
Credit spread	The extra yield over the policy rate that a riskier borrower must pay.
Dilution	The fall in each existing shareholder's stake when a company issues new shares.
Haircut	The share of a debt that creditors give up when a company is restructured.
Net worth	Everything the company owns, at market value, minus what it owes.